LECTURE 1

Databases and Database Users

Dr. Mohamed Maher

AGENDA

Outline

1. Types of Databases and Database Applications
2. Basic Definitions
3. Typical DBMS Functionality
4. Example of a Database (UNIVERSITY)
5. Main Characteristics of the Database Approach
6. Types of Database Users
7. Advantages of Using the Database Approach
8. Historical Development of Database Technology
9. Extending Database Capabilities
10. When Not to Use Databases

FOUNDATIONS

Types of Databases & Database Applications

Traditional Applications

Numeric and textual databases — structured records such as customer, inventory, and financial data.

Our course focuses on traditional applications.

More Recent Applications

- Multimedia databases
- Geographic Information Systems (GIS)
- Biological and Genome databases
- Data Warehouses
- Mobile databases
- Real-time and Active databases
- Big Data

CONTEXT

Recent Developments in Data Capture

Social Networks

Platforms started capturing vast information about people and their communications — posts, tweets, photos, and videos — across networks such as Facebook, Twitter, and LinkedIn. All of this constitutes data.

Search Engines

Google, Bing, and Yahoo collect and maintain their own repository of web pages, built purely to serve fast, relevant searches.

VOCABULARY

Basic Definitions

Database	A collection of related data.
Data	Known facts that can be recorded and have an implicit meaning.
Mini-world (Universe of Discourse)	The part of the real world described by the database — e.g., student grades and transcripts at a university. Changes to the mini-world are reflected in the database.
Database Management System (DBMS)	A software package/system that facilitates the creation and maintenance of a computerized database.
Database System	The DBMS software together with the data itself — sometimes the applications are included too.

INTERACTING WITH DATA

Working with a Database

1 Sharing

Allows multiple users and programs to access the database simultaneously.

2 Application Program

Accesses the database by sending queries to the DBMS.

3 Query

Causes data to be retrieved; part of accessing different parts of data to formulate a result.

4 Transaction

May read data and update values, or generate and store new data in the database.

BEYOND THE BASICS

Additional DBMS Functionality

Protection

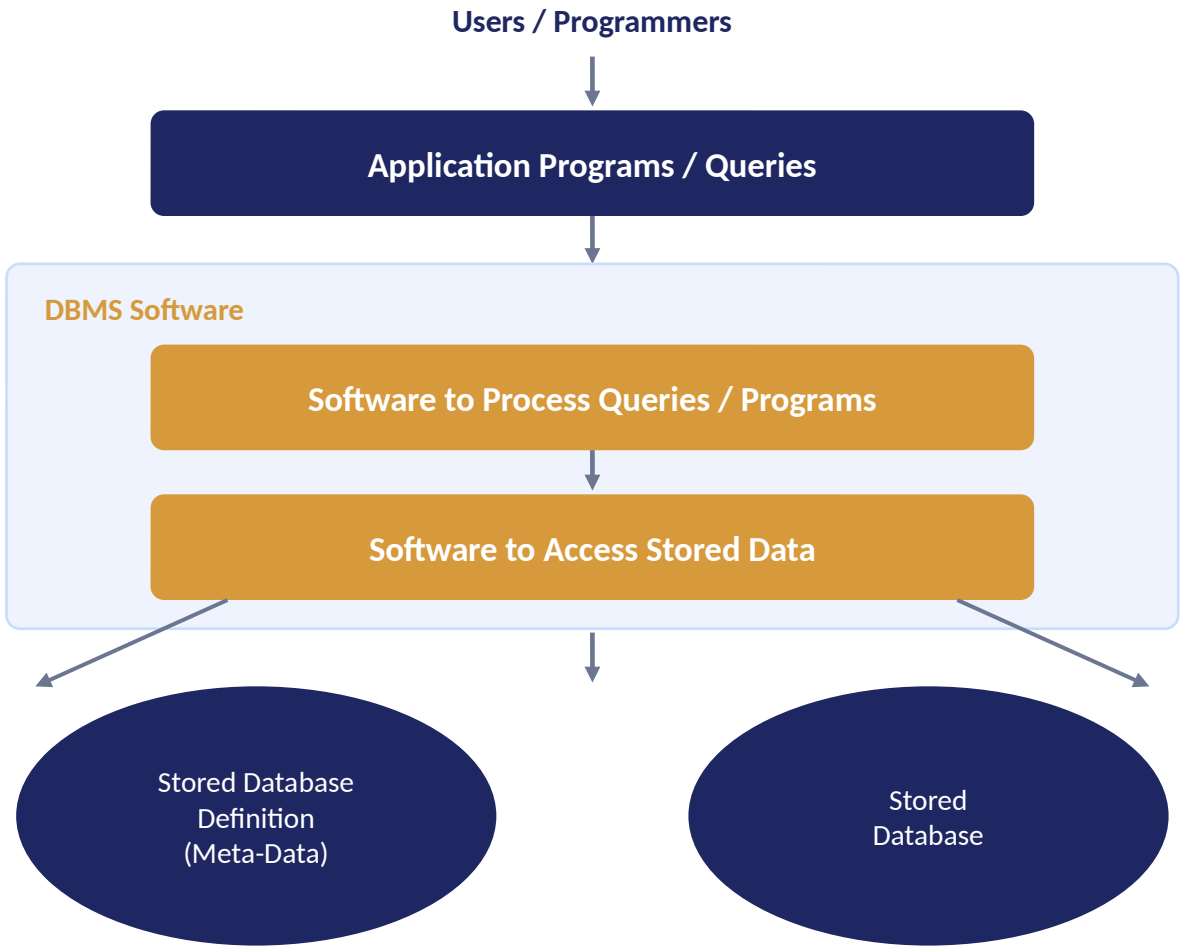
- System protection — against hardware or software malfunction (or crashes)
- Security protection — against unauthorized or malicious access

Maintaining the Database System

- Allows the system to evolve as requirements change over time
- Applications must keep up with changing user requirements against the database

ARCHITECTURE

Simplified Database System Environment



EXAMPLE OF A DATABASE

The UNIVERSITY Mini-World: Entities

Mini-world: part of a UNIVERSITY environment. The database captures the following mini-world entities:



EXAMPLE OF A DATABASE

The UNIVERSITY Mini-World: Relationships

SECTIONS *are of specific* **COURSES**

STUDENTS *take* **SECTIONS**

COURSES *have prerequisite* **COURSES**

INSTRUCTORS *teach* **SECTIONS**

COURSES *are offered by* **DEPARTMENTS**

STUDENTS *major in* **DEPARTMENTS**

These entities and relationships are typically expressed with a conceptual data model, such as the Entity-Relationship (ER) model.

EXAMPLE OF A DATABASE

A Database that Stores Student & Course Data

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to CS	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Math	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_id	Course_no	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang

GRADE_REPORT

Student_number	Section_id	Grade
17	112	B
17	119	C
8	85	A
8	92	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Main Characteristics (Part 1)

1 Self-Describing Nature

A DBMS catalog stores the description of a particular database — its data structures, types, and constraints. This description is called meta-data, allowing the DBMS software to work with different database applications.

2 Program-Data Independence

The structure of data files is stored in the DBMS catalog, separately from the access programs — so data structures and storage organization can change without changing the DBMS access programs.

3 Data Abstraction

A data model hides storage details and presents users with a conceptual representation of the data. Programs refer to data model constructs rather than storage details.

Main Characteristics (Part 2)

4 Multiple Views of the Data

Each user may see a different view of the database, describing only the data of interest to that user.

5 Program-Operation Independence

Operations are specified in two parts: an interface (operation name and argument types) and an implementation, which can change without affecting the interface.

6 Sharing Data & Multi-User Transaction Processing

- Allows a set of concurrent users to retrieve from and update the database
- Concurrency control guarantees each transaction is correctly executed or aborted
- Recovery ensures each completed transaction's effect is permanently recorded
- OLTP (Online Transaction Processing) enables hundreds of concurrent transactions per second

META-DATA IN ACTION

Example of a Simplified Database Catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
...	...	...
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is an enumerated type with all known majors. XXXXNNNN defines a type with four alpha characters followed by four digits.

DATABASE USERS

Who Works with a Database?

Actors on the Scene

Those who actually use and control database content, and those who design, develop, and maintain database applications.

Workers Behind the Scene

Those who design and develop the DBMS software and related tools, and the computer systems operators.

ACTORS ON THE SCENE

Database Administrators & Designers

1 Database Administrators (DBA)

Responsible for authorizing access to the database, coordinating and monitoring its use, acquiring software and hardware resources, controlling its use, and monitoring the efficiency of operations.

2 Database Designers

Responsible for defining the content, structure, constraints, and functions or transactions against the database. They must communicate with end-users and understand their needs.

ACTORS ON THE SCENE

Database End Users

1 Casual

Access the database occasionally, only when needed.

2 Naïve / Parametric

The largest end-user population — they run well-defined “canned transactions.” Mobile app users, bank tellers, and reservation clerks fall here; social media users post and read via canned interfaces.

3 Sophisticated

Business analysts, scientists, and engineers thoroughly familiar with system capabilities — many use tools that work closely with the stored database.

4 Stand-Alone

Maintain personal databases using ready-to-use packaged applications — e.g., a tax program or a personal photo/video library.

ACTORS ON THE SCENE

System Analysts & Application Developers

This category currently accounts for a very large proportion of the IT work force.

1 System Analysts

Understand the requirements of naïve and sophisticated users, and design applications — including canned transactions — to meet those requirements.

2 Application Programmers

Implement the specifications developed by analysts, then test and debug them before deployment.

3 Business Analysts

A growing role: analyzing vast amounts of business and real-time data (“Big Data”) for better decisions in planning, advertising, and marketing.

WORKERS BEHIND THE SCENE

Building & Running the DBMS

1 System Designers & Implementors

Design and implement DBMS packages as modules and interfaces, then test and debug them. The DBMS must interface with applications, language compilers, and operating system components.

2 Tool Developers

Design and implement tools for modeling and designing databases, performance monitoring, prototyping, test-data generation, UI creation, and simulation.

3 Operators & Maintenance Personnel

Manage the actual running and maintenance of the database system's hardware and software environment.

Advantages of the Database Approach (1/3)

1 Controlling Redundancy

Each logical data item — such as a student's name or birth date — is stored in only one place in the database. This is known as data normalization; it ensures consistency and saves storage space.

2 Restricting Unauthorized Access

Only DBA staff use privileged commands and facilities, keeping sensitive data protected from unauthorized users.

Advantages of the Database Approach (2/3)

1 Persistent Storage for Objects

Object-oriented DBMSs, for example, make program objects persistent.

2 Efficient Query Processing

The query processing and optimization module chooses an efficient execution plan for each query based on existing storage structures, such as indexes.

3 Backup & Recovery Services

Ensures the database can be restored and completed transactions are never lost.

4 Multiple User Interfaces

Supports a range of interfaces suited to casual, naïve, and sophisticated users alike.

Advantages of the Database Approach (3/3)

1 Representing Complex Relationships

Naturally models rich relationships among many kinds of data.

2 Enforcing Integrity Constraints

Rules built into the database keep stored data valid and consistent.

3 Deductive & Active Rules

Draws inferences and triggers actions automatically from stored data.

THE BIGGER PICTURE

Additional Implications of the Database Approach

1 Enforcing Standards

Crucial for large organizations — covering data item names, display formats, screens, report structures, meta-data, and web page layouts.

2 Reduced Development Time

The incremental time to add each new application is reduced.

3 Flexible Data Structures

The database structure may evolve as new requirements are defined.

4 Current Information

Extremely important for on-line transaction systems — shopping, airline, hotel, and car reservations.

5 Economies of Scale

Consolidating data and applications across departments avoids wasteful overlap of resources and personnel.

LECTURE SUMMARY

What We Covered Today

1. Types of Databases and Database Applications
2. Basic Definitions
3. Typical DBMS Functionality
4. Example of a Database (UNIVERSITY)
5. Main Characteristics of the Database Approach
6. Types of Database Users
7. Advantages of Using the Database Approach
8. Historical Development of Database Technology
9. Extending Database Capabilities
10. When Not to Use Databases